DETAILED DIAGNOSTIC REPORT

COMPREHENSIVE PROPERTY ANALYSIS & ASSESSMENT

■ AI-Powered Professional Analysis

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Professional Disclaimer: This premium report has been generated using advanced AI analysis of inspection and thermal reports. While our AI system provides comprehensive insights, all findings should be verified by qualified professionals before taking any action. This report represents a sophisticated analysis tool to support expert decision-making.

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**Page numbers are approximate and may vary based on content length*

PROPERTY ISSUE SUMMARY

A comprehensive diagnostic assessment of the residential building situated in Kolkata reveals several critical structural and environmental anomalies requiring immediate attention. The primary concerns identified through a dual-methodology approach, incorporating both visual inspection and advanced thermal imaging, include significant moisture ingress within the Bedroom area, manifesting as visible water leakage and active mold growth, further substantiated by indicative thermal anomalies. Concurrently, the Living Room exhibits evidence of structural distress and compromised thermal envelope performance, characterized by visible surface cracking, paint degradation, and substantial temperature variations suggestive of heat leakage. These findings collectively indicate a potential compromise in the building's integrity and indoor environmental quality, necessitating a detailed and targeted remediation strategy.

AREA-WISE OBSERVATIONS

The assessment identified distinct issues within specific areas of the property. In the Living Room, visual inspection confirmed the presence of visible cracks on the wall

surfaces, accompanied by observed paint peeling. Complementary thermal imaging analysis in this area revealed high temperature variations and a strong indication of possible heat leakage, suggesting deficiencies in the thermal envelope or insulation integrity. Moving to the Bedroom, a direct visual observation of water leakage on the ceiling was made, coupled with the detection of active mold growth. This visual evidence of moisture intrusion was further corroborated by thermal imaging, which identified localized cool spots, unequivocally indicating the presence of moisture, and a broader thermal anomaly suggesting an underlying seepage issue. The convergence of visual and thermal data provides a robust understanding of the conditions present in each space.

PROBABLE ROOT CAUSE

Not Available

THE OBSERVED CONDITIONS INDICATE DISTINCT PROBABLE ROOT CAUSES FOR THE IDENTIFIED ISSUES. IN THE LIVING ROOM, THE COEXISTENCE OF VISIBLE WALL CRACKS, PAINT PEELING, AND THE HIGH TEMPERATURE VARIATIONS WITH SUSPECTED HEAT LEAKAGE, AS DETECTED BY THERMAL IMAGING, STRONGLY SUGGESTS AN UNDERLYING STRUCTURAL MOVEMENT OR SETTLEMENT, POTENTIALLY EXACERBATED BY DEFICIENCIES IN THE BUILDING'S THERMAL ENVELOPE. THESE THERMAL INEFFICIENCIES LIKELY CONTRIBUTE TO INCREASED ENERGY CONSUMPTION AND DISCOMFORT, WHILE THE

CRACKING MAY INDICATE FOUNDATIONAL SHIFTS OR MATERIAL FATIGUE. FOR THE BEDROOM, THE VISIBLE WATER LEAKAGE ON THE CEILING AND THE PRESENCE OF MOLD GROWTH, DIRECTLY CORRELATED WITH THERMAL IMAGING REVEALING COOL SPOTS AND SEEPAGE ANOMALIES, CONCLUSIVELY POINTS TO AN ACTIVE MOISTURE INTRUSION PATHWAY. THIS MOISTURE INGRESS IS THE DIRECT DRIVER FOR BOTH THE VISIBLE WATER DAMAGE AND THE BIOLOGICAL GROWTH, INDICATING A FAILURE IN THE ROOF SYSTEM, PLUMBING, OR EXTERIOR ENVELOPE INTEGRITY ALLOWING WATER PENETRATION.

Not Available

SEVERITY ASSESSMENT

The severity of the identified issues ranges from Medium to High, demanding prompt intervention. The conditions within the Living Room are assessed as Medium severity. While visible cracks and paint peeling are concerning from both aesthetic and structural perspectives, their direct impact on immediate habitability is less critical than active water intrusion. However, the accompanying thermal leakage signifies ongoing energy waste and potential for accelerated material degradation if left unaddressed. The issues detected in the Bedroom, characterized by active water leakage on the ceiling, visible mold growth, and confirmed moisture presence through thermal analysis, are categorized as High severity. Active water

intrusion poses a direct threat to structural components, interior finishes, and electrical systems, while mold growth presents significant health risks to occupants and can rapidly proliferate, causing extensive damage and requiring specialized remediation. The combination of structural damage potential and immediate health implications elevates the Bedroom issues to a critical concern.

RECOMMENDED ACTIONS

To mitigate the identified deficiencies and restore the property to optimal condition, a series of prioritized actions are recommended. For the Living Room, a qualified structural engineer should be engaged to thoroughly assess the extent and origin of the wall cracks, providing recommendations for structural stabilization or repair. Concurrently, a building envelope specialist should investigate the source of the thermal variations and heat leakage, prescribing appropriate insulation and sealing improvements. In the Bedroom, immediate and urgent action is required. A licensed roofing contractor or plumbing professional must promptly identify and repair the source of the water leakage to prevent further ingress and structural damage. Following the complete cessation of water entry and drying of affected materials, a certified mold remediation specialist must be engaged to safely and effectively remove all detected mold growth and contaminated materials, ensuring proper containment and air quality management during the process. Post-remediation, a building professional should oversee the repair and restoration of all damaged ceiling and wall finishes.

ADDITIONAL NOTES

The synergistic application of visual inspection and thermal imaging has proven invaluable in accurately diagnosing both overt and concealed issues within the property,

providing a comprehensive understanding of the underlying conditions. The presence of mold indicates an ongoing and sustained moisture problem that has likely existed for some time, highlighting the importance of immediate moisture source identification and elimination. While specific building codes were not referenced in the provided data, all remedial actions should strictly adhere to local building regulations and best industry practices to ensure the long-term safety and durability of the repairs. Proactive monitoring following remediation is highly advised to confirm the efficacy of repairs and prevent recurrence.

MISSING OR UNCLEAR INFORMATION

To further enhance the precision of this diagnostic report and inform targeted remediation strategies, several data points are currently not available. Specific measurements regarding the dimensions or depth of the wall cracks in the Living Room would be beneficial for a detailed structural assessment. Similarly, quantitative data on the extent of paint peeling and the precise locations and temperatures associated with the high thermal variations and heat leakage would provide a more granular understanding of envelope performance. For the Bedroom, details regarding the exact volume or frequency of water leakage, the precise species and coverage area of mold growth, and quantitative moisture readings of affected materials are absent. Obtaining this additional detailed information would allow for a more precise scope of work for contractors and a more accurate projection of remediation requirements and associated costs.

EXTRACTED IMAGES

Inspection Report Images:

Page 1: inspection_page_1_img_1.jpeg

Thermal Report Images:

Page 1: thermal_page_1_img_1.jpeg

APPENDIX

■ Premium Report Generation Information

This sophisticated Detailed Diagnostic Report has been generated using cutting-edge AI technology combined with advanced building science principles. Our system represents the pinnacle of automated property analysis, delivering insights that traditionally required extensive manual expertise and time investment.

■ Advanced Technical Specifications

- **Analysis Engine:** Google Gemini 2.5 Flash - State-of-the-art AI model
- **PDF Processing:** PyMuPDF Advanced - High-fidelity document extraction
- **Report Generation:** Professional ReportLab Engine - Premium document formatting
- **Processing Date:** 2026-03-19 08:10:23
- **Report Version:** DDR Premium v2.0
- **Quality Assurance:** Multi-layer validation and verification

■ Our Commitment to Excellence

Every report generated through our system undergoes rigorous quality checks and validation processes. We combine the power of artificial intelligence with proven building science methodologies to deliver reports that professionals can trust and clients can understand.

■ Data Processing Capabilities

Our system processes multiple data streams simultaneously, including visual inspection data, thermal imaging information, and environmental factors. This comprehensive approach ensures that no critical information is overlooked and that all findings are properly contextualized within the broader property assessment framework.

■ Continuous Improvement

Our AI models are continuously trained on the latest building science research, industry standards, and real-world inspection data. This ensures that every report reflects the most current understanding of building performance, material science, and diagnostic methodologies.

Thank you for choosing our premium DDR analysis service. We are committed to providing the highest quality property assessments available in the industry today.

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